

# GOSFORD HIGH SCHOOL



## *MATHEMATICS*

### HSC

2009/2010

## Assessment Task 1

*Time Allowed: 70 minutes*

### Instructions:

- All questions are **NOT** of equal value
- Start each **QUESTION** on a new page
- Write on one side of the page only
- Write your name on every page
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used in all sections

**Question 1: (12 Marks)** Start a New Page for this Question

a) Find  $\frac{d}{dx}(1 + 6x^2)^2$  1 Mark

b) Find  $\frac{dy}{dx}$  if  $y = \frac{3x+4}{1-x^3}$  2 Marks

c) Differentiate  $x^3(3 - 2x)^3$ . Write answer in simplest factored form 3 Marks

d) Differentiate  $2x^3 - \frac{5}{x} - 3 - 2\sqrt{x}$  2 Marks

e) i) Calculate: 2 Marks

$$\lim_{x \rightarrow -2} \frac{6x^2 + 21x + 18}{x + 2}$$

ii) A curve  $y = f(x)$  is defined as 2 Marks

$$f(x) = \begin{cases} \frac{6x^2 + 21x + 18}{x + 2} & \text{for } x < -2 \\ x^2 - 1 & \text{for } x \geq -2 \end{cases}$$

Is the curve continuous at  $x = -2$ ? Give reasons for your answer.

**Question 2: (11 Marks)** Start a New Page for this Question

a) Find the centre and radius of a circle with equation  $x^2 - 2x + y^2 + 6y + 9 = 0$  3 Marks

b) A and B are the points  $(-2,0)$  and  $(0,-2)$  respectively. Find the equation of the locus of the point  $P(x,y)$  which moves so that the distance from A to P is three times the distance from B to the point P. 3 Marks

c) For the points A  $(2,-1)$  and B  $(0,1)$  show that the equation of the locus of a point P which moves so that PA and PB are at right angles is  $x^2 - 2x + y^2 = 1$  2 Marks

d) Find the equation of the locus of a point P  $(x,y)$  which moves so that it is equidistant from the point  $(-2,2)$  and the line  $x = 2$  3 Marks

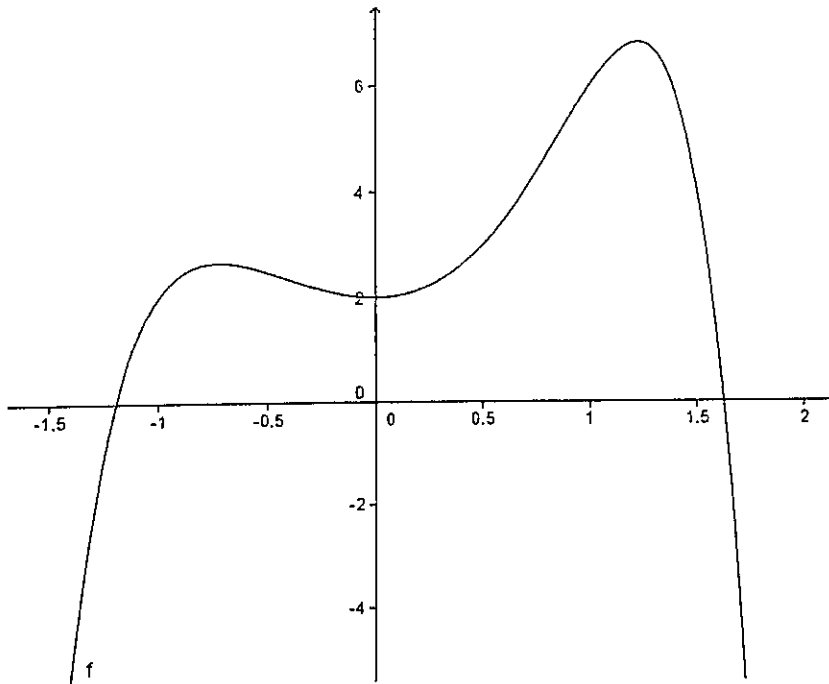
**Question 3: (12 Marks)** Start a New Page for this Question

- a) For the parabola  $y = 4x^2$ , find:
- i) the equation of the directrix 2 Marks
  - ii) the coordinates of the focus 1 Mark
- b) For the parabola  $y = -\frac{1}{2}(x + 2)^2 + 1$
- i) Find the coordinates of the vertex 2 Marks
  - ii) Find the coordinates of the focus 2 Marks
- c) Write down the equation of the parabola with focus at (2,4) and directrix  $y = -2$ . 2 Marks
- d) Find the focal length of the parabola with axis of symmetry  $x = 2$  and vertex (2, -3) which passes through the point (4, -6) 3 Marks

**Question 4: (15 Marks)** Start a New Page for this Question

- a) Consider the curve
- $$y = x^3 - 9x^2 + 15x + 10$$
- i) Find the coordinates of the stationary points on the curve. 3 Marks
  - ii) Determine the nature of each of these stationary points. 2 Marks
  - iii) Find the coordinates of the point of inflexion 2 Marks
  - iv) Neatly sketch the curve for  $0 \leq x \leq 7$ , clearly labelling all important features including the y-intercept. 2 Marks
  - v) What are the greatest and least values of the function in the interval  $0 \leq x \leq 7$  1 Mark
- b) Given  $f(x) = x^3 - 6x^2 + 9x - 5$ , find for what values of  $x$  the function is increasing 2 Marks

- c) Below is a graph of  $y = f(x)$ . Trace or copy the diagram to your answer sheet and on the same number plane neatly sketch  $y = f'(x)$  showing all important features 3 Marks



**Question 5: (11 Marks)** Start a New Page for this Question

- a) At what  $x$ -values on the curve  $y = x^3 - 6x^2 - 30x + 5$  is the tangent to the curve parallel to the line  $y = 6x - 5$  2 Marks
- b) A continuous curve  $y = f(x)$  has the following properties for the interval  $a \leq x \leq b$ : 3 Marks  
 $f(x) > 0$ ,  $f'(x) < 0$  and  $f''(x) > 0$   
 Sketch a curve satisfying these conditions and state the greatest value of  $f(x)$  in this interval.
- c) A quantity ( $Q$ ) is decreasing at a decreasing rate. 2 Marks  
 Describe the signs of  $\frac{dQ}{dt}$  and  $\frac{d^2Q}{dt^2}$  where  $t$  is time
- d) i) Show that the curve  $f(x) = \sqrt{1 - 2x}$  has no stationary points 2 Marks
- ii) Determine the concavity of  $f(x)$  for all  $x$  values 2 Marks

# Assessment Task 1

1) a)  $2(1+6x) \times 12x = 24x(1+6x)$

b)  $y' = \frac{(1-x^3) \times 3 - (3x+4) \times -3x^2}{(1-x^3)^2}$

$$= \frac{3-3x^3+9x^3+12x^2}{(1-x^3)^2}$$

$$= \frac{6x^3+12x^2+3}{(1-x^3)^2}$$

c)  $x^3 \times 3(3-2x) \times 2 + (3-2x) \times 3x^2$   
 $= (3-2x)^2 (-6x^3 + 9x^2 - 6x^3)$   
 $= 3x^2(3-2x)^2(3-4x)$

d)  $y = 2x^3 - 5x^{-1} - 3 - 2x^{-1/2}$   
 $y' = 6x^2 + 5x^{-2} - x^{-1/2}$   
 $y' = 6x^2 + \frac{5}{x^2} - \frac{1}{\sqrt{x}}$

e) i)  $\lim_{x \rightarrow -2} \frac{3(2x^2+7x+6)}{x+2}$   
 $= \lim_{x \rightarrow -2} \frac{3(2x+3)(x+2)}{x+2}$   
 $= 3(2x+3)$   
 $= -3$

ii)  $f(-2) = (-2)^2 - 1$   
 $= 3$

Since  $f(-2) \neq \lim_{x \rightarrow -2} f(x)$  the function is not continuous

2) a)  $x^2 - 2x + 1 + y^2 + 6y + 9 = 0$   
 $(x-1)^2 + (y+3)^2 = 1$

centre:  $(1, -3)$

radius: 1

b)  $PA = \sqrt{(x+2)^2 + y^2}$   
 $PB = \sqrt{x^2 + (y+2)^2}$

$3PB = PA$

$9(x^2 + (y+2)^2) = (x+2)^2 + y^2$

$9x^2 + 9y^2 + 36y + 36 = x^2 + 4x + 4 + y^2$   
 $8x^2 - 4x + 8y^2 + 36y = -32$

c)  $m_{PA} = \frac{y+1}{x-2}$   $m_{PB} = \frac{y-1}{x}$   
 For a right angle

$m_1 m_2 = -1$

$\frac{y+1}{x-2} \times \frac{y-1}{x} = -1$

$y^2 - 1 = -x^2 + 2x$

$x^2 - 2x + y^2 = 1$

d) focus:  $(-2, 2)$

focal length = 2

vertex:  $(0, 2)$

$(y-2)^2 = 4a(x-0)$

$(y-2)^2 = -8x$

3) a) i) focal length =  $\frac{1}{16}$

$y = -\frac{1}{16}$

ii) focus =  $(0, \frac{1}{16})$

b) i)  $(-2, 1)$

ii)  $2(y-1) = (x+2)^2$

$a = \frac{1}{2}$

focus =  $(-2, \frac{1}{2})$

c) focal length = 3

$(x-2)^2 = 12(y-1)$

d)  $(x-2)^2 = 4a(y+3)$

$(4-2)^2 = 4a(-6+3)$

$4 = -12a$

$a = -\frac{1}{3}$

$(x-2)^2 = -\frac{4}{3}(y+3)$

4) a) i)  $y = x^3 - 9x^2 + 15x + 10$

$y' = 3x^2 - 18x + 15$

$y'' = 6x - 18$

$3(x^2 - 6x + 5) = 0$

$(x-5)(x-1) = 0$

$x=1$   $x=5$

$y=17$   $y=-15$

ii)  $(1, 17)$  is a max

$(5, -15)$  is a min

iii)  $6x - 18 = 0$

$x=3$

$y=1$

test  $y''$  on both sides

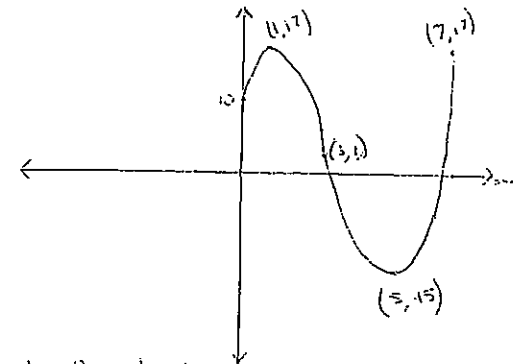
|       |    |   |   |
|-------|----|---|---|
| $x$   | 2  | 3 | 4 |
| $y''$ | -6 | 0 | 6 |

$\therefore (3, 1)$  is a point of inflection

iv) at  $x=0$   $y=10$

$x=7$   $y=17$

y-intercept is 10

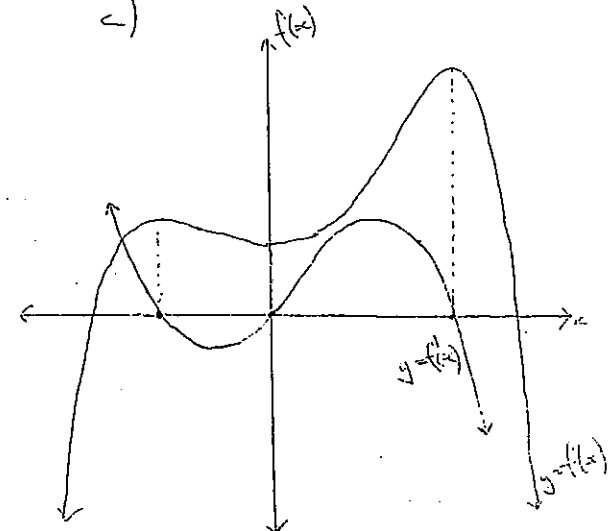


v) Greatest 17

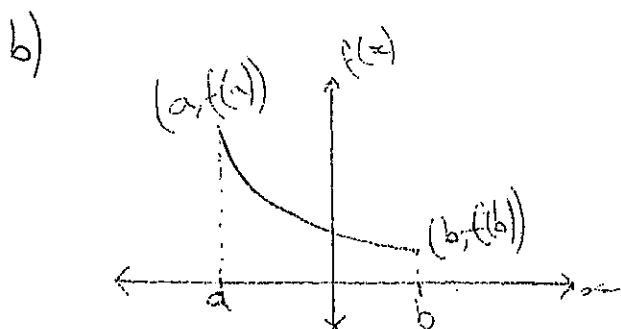
Least -15

b)  $f(x) = 3x^2 - 12x + 9$   
 $3(x^2 - 4x + 3) > 0$   
 $3(x-3)(x-1) > 0$   
 $x < 1$  or  $x > 3$

c)



$$\begin{aligned}
 5) a) \quad y' &= 3x^2 - 12x - 30 \\
 3x^2 - 12x - 30 &= 0 \\
 3(x^2 - 4x - 12) &= 0 \\
 (x-6)(x+2) &= 0 \\
 x &= -2 \text{ or } x = 6
 \end{aligned}$$



Greatest value =  $f(a)$

$$c) \quad \frac{dQ}{dt} < 0 \quad \frac{d^2Q}{dt^2} > 0$$

$$\begin{aligned}
 d) i) \quad f'(x) &= \frac{1}{2}(1-2x)^{-\frac{1}{2}} - 2 \\
 \frac{-1}{\sqrt{1-2x}} &= 0 \\
 -1 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{no stationary points} \\
 ii) \quad f''(x) &= \frac{1}{2}(1-2x)^{-\frac{3}{2}} - 2 \\
 &= -\frac{1}{(\sqrt{1-2x})^3}
 \end{aligned}$$

since  $\sqrt{1-2x}$  is always positive  
 $f''(x)$  is always negative.  
 So  $f(x)$  is concave down

# Question 5

a)  $y = x^3 - 6x^2 - 30x + 5$

where is  $y' = 6$

$$y' = 3x^2 - 12x - 30 = 6$$

$$3x^2 - 12x - 36 = 0$$

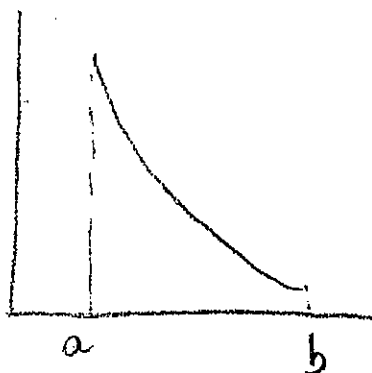
$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \text{ or } -2$$

(2)

b)



Greatest value is  $f(a)$

least value is  $f(b)$

(3)

c)  $\frac{dQ}{dt} < 0$

$\frac{d^2Q}{dt^2} < 0$  (2)

d)  $f(x) = (1-2x)^{\frac{1}{2}}$  , OR  $f'(x) = \frac{1}{2} (1-2x)^{-\frac{1}{2}} \cdot -2 = \frac{-1}{\sqrt{1-2x}}$  (

For stationary pts,  $y' = 0$

$$\frac{-1}{\sqrt{1-2x}} = 0$$

No solutions

ii)  $f''(x) = -1 \cdot -\frac{1}{2} (1-2x)^{-\frac{3}{2}} \cdot -2 = \frac{-1}{\sqrt{(1-2x)^3}}$

$f''(x)$  is always negative in its domain  
Domain  $x \leq \frac{1}{2}$  But  $f''(x)$  undefined at  $x = \frac{1}{2}$

$\therefore$  Curve concave down for  $x < \frac{1}{2}$  (i)  
infinite gradient at  $x = \frac{1}{2}$ . No concavity  
Curve does not exist for  $x > \frac{1}{2}$